

HW 1

Q1



$$Q_1 < Q_2$$

$$\tan Q_2 = \frac{R Q_1}{\frac{R}{2}}$$

(taylor)

$Q_1 < Q_2$ is the condition

$$Q_2 = \frac{R Q_1}{\frac{R}{2}} \} Q_1 < \frac{R Q_1}{\frac{R}{2}}$$

$$\frac{Q_2}{2} < R \} \text{condition}$$

$$mg(R + \frac{a}{2} - R\theta^2 - R + \frac{R\theta^2}{2} - \frac{a}{2} + \frac{a\theta^2}{4})$$

$$mg\frac{\theta^2}{2}(\frac{a}{2} - R) = \frac{1}{2}mv^2 + \frac{1}{2}I\dot{\theta}^2 \quad (\text{Assume the axis is center of cube})$$

$$= \frac{1}{2} \cdot \frac{1}{6}ma^2\dot{\theta}^2$$

$$= \frac{1}{12}ma^2\dot{\theta}^2$$

$$\sqrt{\frac{6g\theta^2(\frac{a}{2} - R)}{a^2}} = \dot{\theta} \rightarrow \theta \cdot \frac{\sqrt{6g(\frac{a}{2} - R)}}{a} = \frac{d\theta}{dt} \rightarrow \int \frac{\sqrt{6g(\frac{a}{2} - R)}}{a} dt = \int \frac{d\theta}{\theta} \rightarrow$$

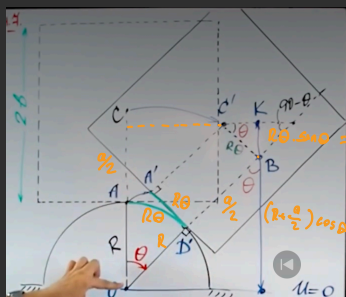
$$\frac{\sqrt{6g(\frac{a}{2} - R)}}{a} \cdot t = \ln \theta \rightarrow \theta = e^{\frac{\sqrt{6g(\frac{a}{2} - R)}}{a} \cdot t} = e^{\frac{\sqrt{6g(R - \frac{a}{2})}}{a} \cdot t} \cdot i \cdot \boxed{e^{i\theta} = \cos \theta + i \sin \theta}$$

$\hookrightarrow \frac{a}{2} < R$ (condition)

$$\theta = \underbrace{\cos \frac{\sqrt{6g(R - \frac{a}{2})}}{a} \cdot t}_{\text{real motion}} + i \sin \frac{\sqrt{6g(R - \frac{a}{2})}}{a} \cdot t \rightarrow \theta = \theta_0 \cos(\omega t)$$

$$\boxed{\omega = \frac{\sqrt{6g(R - \frac{a}{2})}}{a}}$$

I think it is not totally correct since I didn't consider linear velocity.



Q2

$$U(x) = U_0 \cos\left(\frac{x}{x_0}\right) \quad \text{equilibria}$$

$$U'(x) = -\frac{U_0}{x_0} \sin\left(\frac{x}{x_0}\right) \stackrel{!}{=} 0 \rightarrow \sin\left(\frac{x}{x_0}\right) = 0 \rightarrow x = \pi x_0 \cdot c \quad \forall c \in \mathbb{Z}$$

$$U''(x) = \frac{U_0}{x_0^2} \cos\left(\frac{x}{x_0}\right) \quad \begin{array}{l} x = 0 + 2\pi x_0 c \\ x = \pi x_0 + 2\pi x_0 c \end{array} \quad \begin{array}{l} -\frac{U_0}{x_0^2} < 0 \quad \text{unstable} \\ \frac{U_0}{x_0^2} > 0 \quad \text{stable} \end{array}$$

$$U'(x) = F = m\ddot{x} = -\frac{U_0}{x_0} \underbrace{\sin\left(\frac{x}{x_0}\right)}_{\frac{x}{x_0}} \rightarrow \ddot{x} = -\frac{U_0}{m x_0} \cdot \frac{x}{x_0} = -\underbrace{\frac{U_0}{m x_0^2}}_{\omega^2} x \quad \text{by SHO formula}$$

$$\omega = \sqrt{\frac{U_0}{m x_0}} \rightarrow f = \frac{1}{2\pi x_0} \sqrt{\frac{U_0}{m}}$$

$$\underline{x \ll x_0}$$

Q3

$$\vec{X} = \begin{bmatrix} \vec{x}_1 \\ \vec{x}_2 \end{bmatrix} \quad \vec{X} = \begin{bmatrix} \vec{x}_1 \\ \vec{x}_2 \end{bmatrix} \quad \left. \begin{array}{l} \dot{\vec{x}}_1 = 0.1 \hat{x} \text{ m/s} \\ \dot{\vec{x}}_2 = 0 \text{ m/s} \end{array} \right\} t=0$$

$$\left. \begin{array}{l} m \ddot{\vec{x}}_1 = -4k\vec{x}_1 - 0.42k\vec{x}_1 + 0.42k\vec{x}_2 \\ m \ddot{\vec{x}}_2 = -4k\vec{x}_2 - 0.42k\vec{x}_2 + 0.42k\vec{x}_1 \end{array} \right\} \ddot{\vec{X}} = -\omega^2 \begin{bmatrix} 4.42 & -0.42 \\ -0.42 & 4.42 \end{bmatrix} \vec{X}$$

$$\begin{vmatrix} 4.42 - \lambda & -0.42 \\ -0.42 & 4.42 - \lambda \end{vmatrix} = (4.42 - \lambda)^2 - (0.42)^2 = (\lambda - 4)(\lambda - 4.84) = 0 \quad \begin{array}{l} \lambda_1 = 4 \\ \lambda_2 = 4.84 \end{array}$$

for $\lambda_1 = 4$

$$\begin{bmatrix} 0.42 & -0.42 \\ -0.42 & 0.42 \end{bmatrix} \vec{a}_1 = 0 \quad \vec{a}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \text{eigenmode 1}$$

$$-\omega_1^2 = 4 - 4\pi^2 \quad \omega_1 = 4\pi$$

$$\underline{f_1 = 2 \text{ Hz}} \quad \underline{x_1 = x_2}$$

for $\lambda_2 = 4.84$

$$\begin{bmatrix} -0.42 & -0.42 \\ -0.42 & -0.42 \end{bmatrix} \vec{a}_2 = 0 \quad \vec{a}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \quad \text{eigenmode 2}$$

$$-\omega_2^2 = 4.84 - 4\pi^2 \rightarrow \omega_2 = 4.4\pi$$

$$\underline{f_2 = 2.2 \text{ Hz}} \quad \underline{x_1 = -x_2}$$

$$x_1 = A \cos(\omega_1 t + \phi_1) + B \cos(\omega_2 t + \phi_2)$$

A: mode 1

$$x_2 = A \cos(\omega_1 t + \phi_1) - B \cos(\omega_2 t + \phi_2)$$

B: mode 2

$$v_1 = -A\omega_1 \sin(\omega_1 t + \phi_1) - B\omega_2 \sin(\omega_2 t + \phi_2)$$

$$v_2 = -A\omega_1 \sin(\omega_1 t + \phi_1) + B\omega_2 \sin(\omega_2 t + \phi_2)$$

t=0

$$\begin{array}{l} A \cos \phi_1 + B \cos \phi_2 = 0 \quad \phi_1 = \pi/2 \\ A \cos \phi_1 - B \cos \phi_2 = 0 \quad \phi_2 = \pi/2 \end{array}$$

$$\begin{array}{l} v_1(t) = 0.05 \cdot \sin(4\pi t + \pi/2) + 0.05 \sin(4.4\pi t + \pi/2) \\ v_2(t) = 0.05 \cdot \sin(4\pi t + \pi/2) - 0.05 \sin(4.4\pi t + \pi/2) \end{array}$$

$$-A\omega_1 \sin \phi_1 - B\omega_2 \sin \phi_2 = 0.1$$

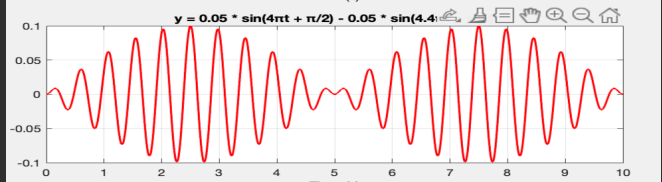
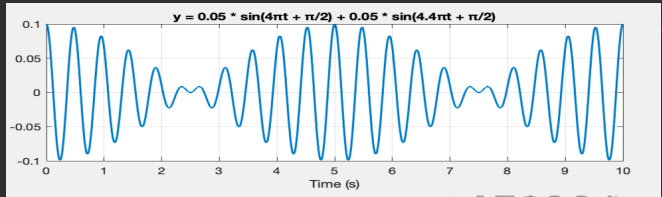
$$-A\omega_1 \sin \phi_1 + B\omega_2 \sin \phi_2 = 0$$

↓

$$A\omega_1 \sin \phi_1 = B\omega_2 \sin \phi_2 = -0.05$$

$$A\omega_1 = B\omega_2 = -0.05$$

$$A 4\pi = B 4.4\pi = -0.05$$



Experimental Project: Coupled Pendulum System

1. Experimental Setup

Two smartphones were hung as simple pendulums from the same string. The string length was adjusted so that each pendulum's oscillation period was close to 1.0 s. Using the relation $T = 2\pi\sqrt{L/g}$ with $g = 9.81 \text{ m/s}^2$ and $T \approx 1.0$, the estimated length was about $L = 0.25 \text{ m}$ approximately.

In practice, the length was fine-tuned by small adjustments until the measured period was around one second.

The phones were attached with light strings to a common horizontal support, allowing mechanical coupling through the shared mounting.

All phones ran the **Phyphox** app to record Y-axis acceleration at roughly 100 Hz.

Two main tests were performed:

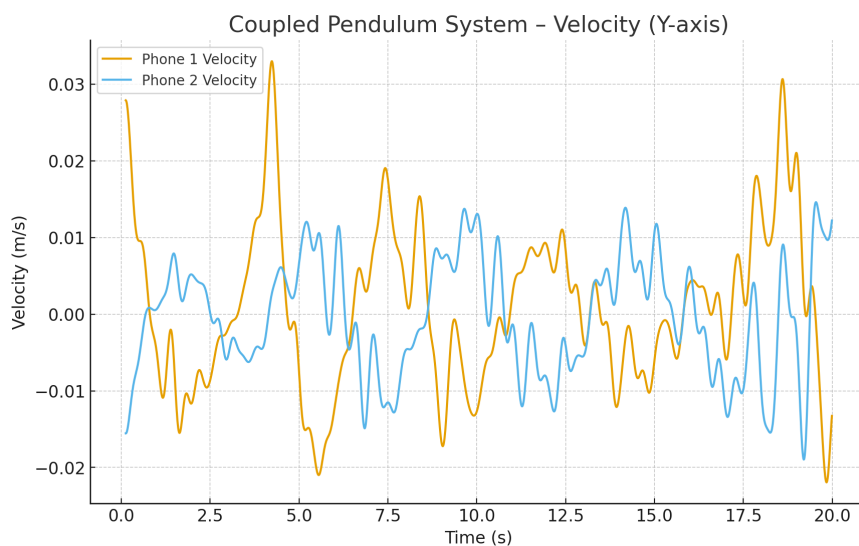
- (a) displacing only one pendulum while keeping the other still, and
- (b) releasing both pendulums together in the same direction.

2. Single Pendulum Perturbation

In this test, only one pendulum was displaced and released while the other remained at rest.

Acceleration and velocity data were filtered between 0.2 and 3 Hz to remove background noise.

Over time, the previously stationary pendulum began to move, showing that energy was transferred through the coupling — just as expected for weakly coupled oscillators.



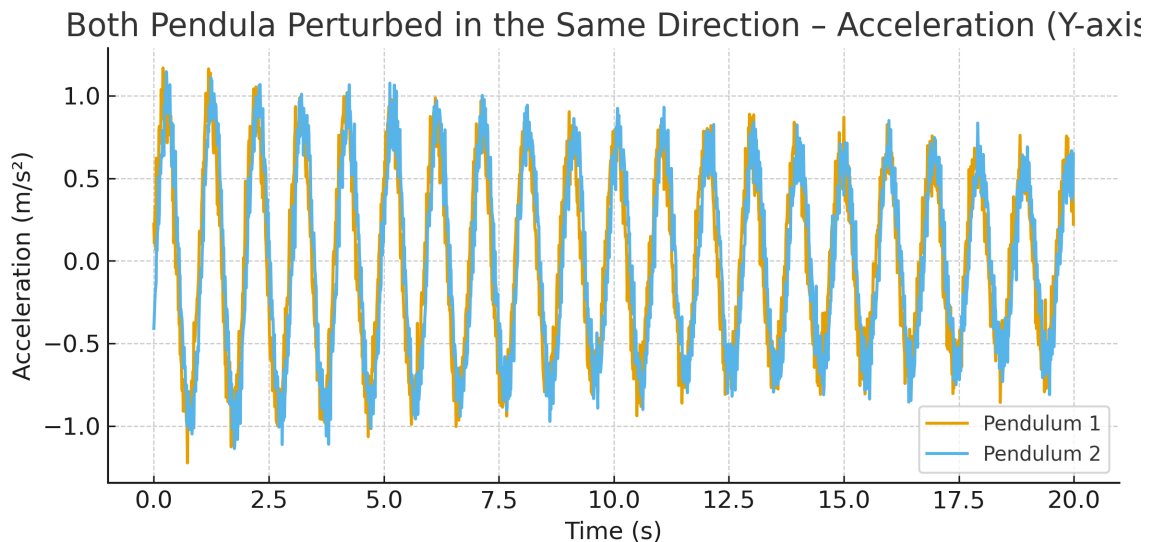
Phase comparison using both the **Hilbert transform** and **peak delay** methods showed an average phase difference of about **43°**, which corresponds to a **0.12 s time lag** for the measured oscillation period.

3. Both Pendula Perturbed in the Same Direction

When both pendulums were released together in the same direction, their motions stayed almost synchronized.

Small differences in amplitude and phase were observed due to minor mechanical imperfections and sensor noise.

This configuration represents the **in-phase normal mode** of the coupled system.



Phase analysis showed a smaller mean phase offset of about **22°**, equal to roughly **0.06 s delay**.

This confirms that both pendulums oscillated nearly in phase and that coupling was weak, allowing only slight energy exchange.

4. Discussion

These two tests highlight how sensitive a coupled system is to its starting conditions.

When only one pendulum was disturbed, energy flowed back and forth, creating visible phase shifts and amplitude changes.

When both were released together, the system stayed synchronized with almost no energy transfer.

This demonstrates how coupled oscillators can switch between out-of-phase and in-phase behavior depending on how they are started.

5. Conclusion

This experiment clearly showed the main properties of coupled oscillators: synchronization, energy transfer, and dependence on initial conditions.

With the pendulum period tuned near one second, both scenarios could be compared directly.

The observations matched theoretical predictions, showing measurable phase differences and energy sharing caused by the mechanical coupling.

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